

Polynomial Extensions Need Not Preserve n -Absorbing Ideals

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Abstract

We resolve an open problem [3] in commutative ring theory and prove that polynomial extensions need not preserve n -absorbing ideals. The proof is generated by GPT 5.5 Pro and verified by Claude Opus 4.8 ultracode.

1 Introduction

We assume that all rings are commutative with identity. Following Anderson and Badawi [1], we define *n -absorbing rings* as follows.

Definition 1.1 ([1]). Let $n \geq 1$ be an integer. A proper ideal I of a ring R is *n -absorbing* if whenever $x_1 \cdots x_{n+1} \in I$ with $x_1, \dots, x_{n+1} \in R$, there exist n of the x_i whose product lies in I . The *absorbing number* of I is

$$\omega_R(I) = \min\{n \geq 1 : I \text{ is } n\text{-absorbing}\} \in \{1, 2, \dots\} \cup \{\infty\}.$$

A 1-absorbing ideal is exactly a prime ideal, and a 2-absorbing ideal is the notion of Badawi [2]. [4] asks the following question: if I is an n -absorbing ideal of R , is $I[X]$ an n -absorbing ideal of the polynomial ring $R[X]$? Since one always has $\omega_{R[X]}(I[X]) \geq \omega_R(I)$ (specialize to constant polynomials), an equivalent formulation is the Anderson–Badawi conjecture

$$\omega_{R[X]}(I[X]) = \omega_R(I) \quad \text{for every ideal } I \text{ of every ring } R. \quad (1.1)$$

The conjecture is affirmative for $n \leq 2$ [1] and has been verified for several classes of rings—Prüfer domains and Gaussian rings with torsion-free additive group, and more generally algebras over an infinite field [5]. To the best of our knowledge, no counterexample had appeared. We refute (1.1) for every $n \geq 3$.

2 Main Proof

Theorem 2.1. *For every integer $n \geq 3$ there are a Noetherian ring A and the ideal $I = (0)$ such that I is n -absorbing in A but $I[X] = (0)$ is not n -absorbing in $A[X]$. Concretely, writing $q = n - 1 \geq 2$, the ring $A = A_q$ of Section 2.2 satisfies*

$$\omega_{A_q}((0)) = q + 1 = n, \quad \omega_{A_q[X]}((0)) = q + 2 = n + 1.$$

The ring A_q is finite as a module over $\mathbb{F}_2[t]$ and may be taken Noetherian local (Remark 2.13). Consequently the Anderson–Badawi conjecture (1.1) and Problem 30(c) have a negative answer for all $n \geq 3$; since they are affirmative for $n \leq 2$, the result is sharp.

The construction is completely explicit. Over $\mathbb{F}_2$ and $D = \mathbb{F}_2[t]$ the ring A_q is a 7-generator D -module carrying a square-free nilpotent layer J , and the failure of n -absorption in $A_q[X]$ is produced by the two linear polynomials

$$f = e_2 + Xe_3, \quad g = (1 + X)e_1 + (X + X^2)e_2 + X^2e_3$$

of Section 2.5, whose product collapses in characteristic two to $fg = X(1 + X)s$, supplying one annihilating factor more than any product of constants can. The argument rests on three elementary facts about A_q :

- (i) its nilpotent part J satisfies $J^3 = 0$ and $A_q/J \cong D$ (Section 2.2);
- (ii) a *characteristic-two obstruction*: a product of two elements of J that lands in the top socle layer is forced to be divisible by t (Lemma 2.7); this pins down $\omega_{A_q}((0)) = q + 1$;
- (iii) over $A_q[X]$ the obstruction *disappears* (its proof needs the residue field $\mathbb{F}_2$, and $\mathbb{F}_2[X]$ is not a field), and the polynomials f, g realize precisely the missing case, giving $\omega_{A_q[X]}((0)) = q + 2$ (Section 2.5).

The finite computations underlying the construction were independently verified by direct symbolic calculation over $\mathbb{F}_2$; see Section 2.7.

2.1 The absorbing number via irredundant zero products

We first record the standard reduction that lets us work with the zero ideal of a quotient.

Lemma 2.2. *Let I be a proper ideal of R and put $A = R/I$. Then for every $n \geq 1$:*

$$I \text{ is } n\text{-absorbing in } R \iff (0) \text{ is } n\text{-absorbing in } A,$$

and the canonical isomorphism $R[X]/I[X] \cong A[X]$ gives

$$I[X] \text{ is } n\text{-absorbing in } R[X] \iff (0) \text{ is } n\text{-absorbing in } A[X].$$

In particular Problem 30(c) for the ideal I is equivalent to the implication

$$(0) \text{ } n\text{-absorbing in } A \implies (0) \text{ } n\text{-absorbing in } A[X].$$

Proof. The defining condition of n -absorption involves only products and membership in the ideal, both of which are detected by the quotient map $R \twoheadrightarrow R/I = A$: a product $x_1 \cdots x_{n+1}$ lies in I iff $\bar{x}_1 \cdots \bar{x}_{n+1} = 0$ in A , and a subproduct lies in I iff the corresponding subproduct vanishes in A . This gives the first equivalence. For the second, $I[X] = \ker(R[X] \rightarrow (R/I)[X])$, so $R[X]/I[X] \cong A[X]$, and the first equivalence applied to $I[X] \subseteq R[X]$ gives the claim. $\square$

Thus it suffices to produce a single ring A in which (0) is n -absorbing but $(0)[X]$ is not; the ideal of the original problem is then $I = (0)$. Fix a nonzero ring A .

Definition 2.3. An *irredundant zero product* in A is a finite sequence $a_1, \dots, a_m \in A$ ($m \geq 1$) with

$$a_1 a_2 \cdots a_m = 0 \quad \text{and} \quad \prod_{i \neq j} a_i \neq 0 \quad \text{for every } j \in \{1, \dots, m\}.$$

(The omitted product for $m = 1$ is the empty product $1 \neq 0$, so a single factor equal to 0 is irredundant of length 1) Let

$$\Omega(A) = \sup\{m : \text{there is an irredundant zero product of length } m\} \in \{1, 2, \dots\} \cup \{\infty\}.$$

The next proposition identifies $\Omega(A)$ with the absorbing number $\omega_A((0))$ of Definition 1.1, reducing the verification of n -absorption to a length bound.

Proposition 2.4. *Let A be a nonzero ring and $n \geq 1$. Then (0) is n -absorbing in A iff every irredundant zero product in A has length $\leq n$. Consequently $\omega_A((0)) = \Omega(A)$.*

Proof. ($\Leftarrow$) Assume every irredundant zero product has length $\leq n$, and let $x_1 \cdots x_{n+1} = 0$. Among all subsets $S \subseteq \{1, \dots, n+1\}$ with $\prod_{i \in S} x_i = 0$ (the full set is one such) choose one of minimal cardinality, say S_0 . Minimality means that omitting any single index of S_0 leaves a nonzero product, so $(x_i)_{i \in S_0}$ is an irredundant zero product; hence $|S_0| \leq n$. Since $|S_0| \leq n < n+1$, we may enlarge S_0 to a set T with $|T| = n$; then $\prod_{i \in T} x_i = (\prod_{i \in S_0} x_i) \cdot \prod_{i \in T \setminus S_0} x_i = 0$. Thus n of the x_i have product 0, so (0) is n -absorbing.

($\Rightarrow$) Assume (0) is n -absorbing and suppose, for contradiction, that $a_1 \cdots a_m = 0$ is irredundant with $m \geq n+1$. Set

$$b_i = a_i \quad (1 \leq i \leq n), \quad b_{n+1} = a_{n+1} a_{n+2} \cdots a_m,$$

so $b_1 \cdots b_{n+1} = a_1 \cdots a_m = 0$. By n -absorption some b_j may be deleted with $\prod_{i \neq j} b_i = 0$.

- If $j \leq n$, then $\prod_{i \neq j} b_i = \prod_{i \in \{1, \dots, m\} \setminus \{j\}} a_i$, which is nonzero by irredundancy of (a_i) — contradiction.
- If $j = n+1$, then $b_1 \cdots b_n = a_1 \cdots a_n = 0$. But then deleting the single index $n+1$ from (a_i) gives $\prod_{i \neq n+1} a_i = (a_1 \cdots a_n)(a_{n+2} \cdots a_m) = 0$, again contradicting irredundancy.

Both cases are impossible, so $m \leq n$.

The two implications show that (0) is n -absorbing iff $\Omega(A) \leq n$, whence $\omega_A((0)) = \min\{n : \Omega(A) \leq n\} = \Omega(A)$. $\square$

In particular, to prove that (0) is n -absorbing it suffices to bound the length of every irredundant zero product by n ; to prove that (0) is *not* n -absorbing it suffices to exhibit one irredundant zero product of length $n+1$.

We will repeatedly use the t -adic valuation. Let B be a ring in which t is a nonzero nonunit and B/tB is a domain; for $0 \neq b \in B$ write $v_t(b) = \max\{k : b \in t^k B\}$ when this is finite, and put $v_t(0) = \infty$. The two cases we need are $B = D = \mathbb{F}_2[t]$ and $B = D[X] = \mathbb{F}_2[t, X]$.

Lemma 2.5. *Let B be a domain in which B/tB is a domain. Suppose $d_1, \dots, d_r \in B$, $r \geq 1$, $k \geq 0$ satisfy*

$$d_1 \cdots d_r \in t^k B \quad \text{and} \quad \prod_{i \neq j} d_i \notin t^k B \quad \text{for every } j.$$

Then $r \leq k$.

Proof. Because B/tB is a domain, v_t is additive: $v_t(bb') = v_t(b) + v_t(b')$ for nonzero b, b' . The first hypothesis gives $\sum_i v_t(d_i) \geq k$, and the second gives $\sum_{i \neq j} v_t(d_i) \leq k-1$ for each j . Subtracting, $v_t(d_j) \geq 1$ for every j . Hence for each j , $k-1 \geq \sum_{i \neq j} v_t(d_i) \geq r-1$, so $r \leq k$. $\square$

2.2 The ring A_q

Fix the polynomial ring $D = \mathbb{F}_2[t]$, and fix an integer $q \geq 2$. Define the D -module

$$A_q = D \cdot 1 \oplus De_1 \oplus De_2 \oplus De_3 \oplus Du_1 \oplus Du_2 \oplus (D/t^q D)s, \quad (2.1)$$

free of rank 6 on $1, e_1, e_2, e_3, u_1, u_2$ together with one torsion generator s of annihilator $t^q D$. Write

$$J = De_1 + De_2 + De_3 + Du_1 + Du_2 + (D/t^q D)s, \quad W = Du_1 + Du_2 + (D/t^q D)s,$$

so that $A_q = D \cdot 1 \oplus J$ and $W \subseteq J$. We turn A_q into a commutative ring by declaring 1 to be the identity and defining a D -bilinear, commutative product on J through the table

$\cdot$	e_1	e_2	e_3	(2.2)
e_1	0	0	$u_2 + s$	
e_2	0	u_2	u_1	
e_3	$u_2 + s$	u_1	u_1	

together with the rule that every product of one of u_1, u_2, s with any element of J is zero. Equivalently, W annihilates J (so $W \cdot J = 0$ and W is a square-zero ideal), and the bilinear extension of (2.2) computes products of e -vectors. The product is well defined on the torsion summand: the only basis products involving s are $e_1 e_3 = u_2 + s$ and the D -scalings $d \cdot s \in (D/t^q D)s$, interpreted in $D/t^q D$, and any product having s as a factor is 0, hence independent of the chosen representative.

Lemma 2.6. *With the product just defined:*

$$J^2 = W, \quad J W = 0, \quad W^2 = 0, \quad J^3 = 0, \quad A_q/J \cong D.$$

Moreover A_q is associative, hence a commutative ring, and it is Noetherian.

Proof. The products. From (2.2), $e_i e_j \in Du_1 + Du_2 + (D/t^q D)s = W$ for all i, j (the values are $0, u_1, u_2, u_2 + s$), and these values generate W over D : indeed $e_2 e_2 = u_2$, $e_2 e_3 = u_1$, and $e_1 e_3 = u_2 + s$ give u_2, u_1 and hence s . Since $W \cdot J = 0$ by construction, $J^2 = W$, $JW = 0$, $W^2 = 0$, and $J^3 = J \cdot J^2 = J \cdot W = 0$. Finally J is an ideal with $A_q/J = D \cdot 1 \cong D$ because $1 \mapsto 1$ and $J \mapsto 0$.

Associativity. Write $a_i = \delta_i \cdot 1 + x_i$ with $\delta_i \in D$ and $x_i \in J$ ($i = 1, 2, 3$). Using D -bilinearity, commutativity of D , and the fact that 1 is a central identity, one expands $(a_1 a_2) a_3$ and $a_1 (a_2 a_3)$ and finds that every term containing at least one scalar factor δ_i agrees in the two expansions (it is a D -multiple of a product of fewer than three J -elements, and module scalar multiplication is associative). Hence $(a_1 a_2) a_3 - a_1 (a_2 a_3) = (x_1 x_2) x_3 - x_1 (x_2 x_3)$. Both terms on the right lie in $J^3 = 0$, so the difference is 0. Thus A_q is associative.

Noetherian. By (2.1), A_q is finitely generated as a module over the Noetherian ring $D = \mathbb{F}_2[t]$. A ring that is module-finite over a Noetherian subring is Noetherian (its ideals are D -submodules of a Noetherian D -module), so A_q is Noetherian. $\square$

For reference, A_q is the quotient of $\mathbb{F}_2[t, e_1, e_2, e_3, u_1, u_2, s]$ by the ideal

$$(e_1^2, e_1 e_2, e_1 e_3 - u_2 - s, e_2^2 - u_2, e_2 e_3 - u_1, e_3^2 - u_1, t^q s) + (u_1, u_2, s) \cdot (e_1, e_2, e_3, u_1, u_2, s),$$

with (2.1) its D -module structure; we use only the intrinsic description. The nilradical of A_q is $\sqrt{(0)} = J$ (nilpotent by $J^3 = 0$, and $A_q/J \cong D$ is reduced), with embedded maximal ideal $\mathfrak{m} = (t, e_1, e_2, e_3, u_1, u_2, s)$ of residue field $A_q/\mathfrak{m} \cong \mathbb{F}_2$; the finiteness of this residue field is what makes the construction possible (Remark 2.14).

2.3 The characteristic-two obstruction

The following lemma is the arithmetic heart of the construction. For $z \in W$ we write its components as $z = \alpha u_1 + \beta u_2 + \gamma s$ with $\alpha, \beta \in D$ and $\gamma \in D/t^q D$.

Lemma 2.7. *Let $x, y \in J$ and suppose $xy \in (D/t^q D)s$, i.e. the u_1 - and u_2 -components of xy vanish. Then $xy \in t(D/t^q D)s$; equivalently, writing $xy = \gamma s$, one has $\gamma \equiv 0 \pmod{t}$.*

Proof. Write $x = ae_1 + be_2 + ce_3 + w$ and $y = a'e_1 + b'e_2 + c'e_3 + w'$ with $a, \dots, c' \in D$ and $w, w' \in W$. Since W annihilates J and $W^2 = 0$, the W -parts w, w' contribute nothing to xy , so by (2.2)

$$xy = (bc' + cb' + cc')u_1 + (ac' + ca' + bb')u_2 + (ac' + ca')s, \quad (2.3)$$

with coefficients in D . (Here $e_1e_3 = u_2 + s$ contributes $(ac' + ca')$ to both the u_2 - and the s -coefficient; $e_2e_2 = u_2$ contributes bb' to u_2 ; $e_2e_3 = u_1$ and $e_3e_3 = u_1$ contribute $bc' + cb'$ and cc' to u_1 ; $e_1e_1 = e_1e_2 = 0$.)

By hypothesis the u_1 - and u_2 -coefficients vanish in D :

$$bc' + cb' + cc' = 0, \quad ac' + ca' + bb' = 0. \quad (2.4)$$

We must show $v_t(ac' + ca') \geq 1$, i.e. that the s -coefficient reduces to 0 modulo t . Reduce (2.4) modulo t , writing $\bar{a} = a \bmod t \in k = \mathbb{F}_2$, etc.:

$$\bar{b}\bar{c}' + \bar{c}\bar{b}' + \bar{c}\bar{c}' = 0, \quad \bar{a}\bar{c}' + \bar{c}\bar{a}' + \bar{b}\bar{b}' = 0 \quad \text{in } \mathbb{F}_2. \quad (2.5)$$

Suppose, for contradiction, that the reduced s -coefficient is nonzero, i.e. $\bar{a}\bar{c}' + \bar{c}\bar{a}' = 1$. The second equation of (2.5) then forces $\bar{b}\bar{b}' = \bar{a}\bar{c}' + \bar{c}\bar{a}' = 1$ (characteristic 2), so $\bar{b} = \bar{b}' = 1$. Substituting $\bar{b} = \bar{b}' = 1$ into the first equation of (2.5) gives $\bar{c} + \bar{c}' + \bar{c}\bar{c}' = 0$ in $\mathbb{F}_2$. Checking the four possibilities $(\bar{c}, \bar{c}') \in \{0, 1\}^2$, the only solution is $\bar{c} = \bar{c}' = 0$. But then $\bar{a}\bar{c}' + \bar{c}\bar{a}' = 0$, contradicting our assumption that it equals 1. Hence $\bar{a}\bar{c}' + \bar{c}\bar{a}' = 0$, i.e. $t \mid ac' + ca'$, as claimed. $\square$

The conclusion reads: *a product of two elements of J that lands in the top socle layer $(D/t^q D)s$ necessarily lands in the smaller layer $t(D/t^q D)s$.* This is a genuine restriction over $\mathbb{F}_2$; Section 2.5 shows it has no analogue over $\mathbb{F}_2[X]$, and that is exactly where the counterexample comes from.

2.4 The absorbing number of A_q

Proposition 2.8. *For every $q \geq 2$, $\omega_{A_q}((0)) = q + 1$. In particular (0) is $(q+1)$ -absorbing in A_q .*

Proof. By Proposition 2.4 it suffices to show $\Omega(A_q) = q + 1$.

Lower bound $\Omega(A_q) \geq q + 1$. Consider the length- $(q+1)$ sequence $s, \underbrace{t, \dots, t}_q$, where t denotes $t \cdot 1 \in A_q$. Its product is $t^q s = 0$. Omitting the factor s leaves $t^q \neq 0$ in $D \cdot 1$. Omitting one of

the q copies of t leaves $t^{q-1}s \neq 0$, since $t^{q-1} \notin t^q D$. Hence the sequence is an irredundant zero product of length $q+1$, and $\Omega(A_q) \geq q+1$.

Upper bound $\Omega(A_q) \leq q+1$. Let $a_1 \cdots a_m = 0$ be an irredundant zero product. We may assume $m \geq 2$ (for $m \leq 1$, $m \leq q+1$ since $q \geq 2$). Write each $a_i = d_i \cdot 1 + x_i$ with $d_i \in D$, $x_i \in J$, and let $h = \#\{i : d_i = 0\}$ be the number of factors lying in J . The images of the a_i in $A_q/J \cong D$ are the d_i ; their product is 0 in the domain D , so some $d_i = 0$ and $h \geq 1$. We bound m according to h .

Case $h \geq 3$. Fix three indices i_1, i_2, i_3 with $a_{i_\ell} = x_{i_\ell} \in J$. If $m \geq 4$, choose $j \notin \{i_1, i_2, i_3\}$; then $\prod_{i \neq j} a_i$ has $a_{i_1} a_{i_2} a_{i_3} = x_{i_1} x_{i_2} x_{i_3} \in J^3 = 0$ as a factor, so it vanishes, contradicting irredundancy. Hence $m \leq 3 \leq q+1$.

Case $h = 2$. Let the two J -factors be $x, y \in J$ and write the remaining $r = m - 2$ factors as $a_i = d_i \cdot 1 + x_i$ with $d_i \neq 0$. Put $z = xy \in J^2 = W$. For any $a = d \cdot 1 + x' \in A_q$ ($x' \in J$) we have $za = dz + zx' = dz$, because $zx' \in WJ = 0$; iterating, the total product equals

$$a_1 \cdots a_m = (d_1 \cdots d_r) z. \quad (2.6)$$

If $r = 0$ (so $m = 2$) then $z = xy = 0$ and irredundancy gives $x, y \neq 0$; here $m = 2 \leq q+1$. Assume $r \geq 1$ and set $d = d_1 \cdots d_r \neq 0$. If $z = 0$, then deleting any one of the non- J factors still leaves a product $(\prod_{i \neq j} d_i)z = 0$, contradicting irredundancy; hence $z \neq 0$. Write $z = \alpha u_1 + \beta u_2 + \gamma s$. From (2.6), $0 = dz = d\alpha u_1 + d\beta u_2 + d\gamma s$. As $Du_1 \oplus Du_2$ is free and $d \neq 0$, this forces $\alpha = \beta = 0$, so $0 \neq z = \gamma s \in (D/t^q D)s$ with $\gamma \neq 0$. Now $z = xy$ lies in $(D/t^q D)s$, so Lemma 2.7 gives $v := v_t(\gamma) \geq 1$ (and $v < q$ since $\gamma \neq 0$ in $D/t^q D$). The annihilator of $z = \gamma s$ in D is

$$(0 :_D \gamma s) = \{e \in D : e\gamma \in t^q D\} = t^{q-v} D, \quad (2.7)$$

because $\gamma = t^v \gamma_0$ with γ_0 a unit modulo t , so $e\gamma \in t^q D$ iff $t^{q-v} \mid e$. By (2.6) and $dz = 0$ we get $d = d_1 \cdots d_r \in t^{q-v} D$; and by irredundancy, deleting any non- J factor a_j leaves $(\prod_{i \neq j} d_i)z \neq 0$, i.e. $\prod_{i \neq j} d_i \notin (0 :_D z) = t^{q-v} D$. Lemma 2.5 (with $B = D$, $k = q - v$) yields $r \leq q - v$, hence $m = 2 + r \leq 2 + (q - v) \leq 2 + (q - 1) = q + 1$.

Case $h = 1$. Let $x \in J$ be the unique J -factor and write the other $r = m - 1$ factors as $a_i = d_i \cdot 1 + x_i$ with $d_i \neq 0$. Working modulo the ideal $W = J^2$, and using $xx_i \in J^2 = W$, $a_1 \cdots a_m \equiv (d_1 \cdots d_r)x \pmod{W}$. The left side is 0, so $(d_1 \cdots d_r)\bar{x} = 0$ in $J/W \cong De_1 \oplus De_2 \oplus De_3 \cong D^3$. Since D^3 is torsion-free and $d_1 \cdots d_r \neq 0$, we get $\bar{x} = 0$, i.e. $x \in W$. Now $x \in W$ gives $xx_i = 0$, so the total product is exactly $(d_1 \cdots d_r)x$. Writing $x = \alpha u_1 + \beta u_2 + \gamma s$, the same freeness argument forces $\alpha = \beta = 0$ and $x = \gamma s$ with $\gamma \neq 0$ (as $x \neq 0$). Here x need not be a product of two J -elements, so the obstruction does not apply and we only know $v = v_t(\gamma) \geq 0$. By (2.7), $(0 :_D x) = t^{q-v} D$, and irredundancy together with Lemma 2.5 gives $r \leq q - v \leq q$. Hence $m = 1 + r \leq q + 1$.

In every case $m \leq q + 1$, so $\Omega(A_q) \leq q + 1$. Combined with the lower bound, $\Omega(A_q) = \omega_{A_q}((0)) = q + 1$. $\square$

2.5 Polynomial cancellation: $(0)[X]$ is not $(q+1)$ -absorbing

We now pass to $A_q[X]$. As a $D[X]$ -module, $A_q[X] = D[X] \cdot 1 \oplus J[X]$, with $J[X]^2 = W[X]$, $J[X]W[X] = 0$, $J[X]^3 = 0$, and $A_q[X]/J[X] \cong D[X]$, a domain; here $W[X] = D[X]u_1 + D[X]u_2 + (D/t^q D)[X]s$. Define

$$f = e_2 + Xe_3 \in J[X], \quad g = (1 + X)e_1 + (X + X^2)e_2 + X^2e_3 \in J[X]. \quad (2.8)$$

Lemma 2.9. *In $A_q[X]$ one has $fg = X(1 + X)s$.*

Proof. Expand $fg = e_2g + Xe_3g$ using (2.2). First

$$e_2g = (1 + X)\underbrace{e_2e_1}_0 + (X + X^2)\underbrace{e_2e_2}_{u_2} + X^2\underbrace{e_2e_3}_{u_1} = (X + X^2)u_2 + X^2u_1.$$

Next, using $e_3e_1 = u_2 + s$, $e_3e_2 = u_1$, $e_3e_3 = u_1$,

$$e_3g = (1 + X)(u_2 + s) + (X + X^2)u_1 + X^2u_1,$$

which, since $X + X^2 + X^2 = X$ in characteristic 2, equals $e_3g = (1 + X)u_2 + (1 + X)s + Xu_1$. Therefore $Xe_3g = X(1 + X)u_2 + X(1 + X)s + X^2u_1$, and adding,

$$fg = (X^2 + X^2)u_1 + ((X + X^2) + X(1 + X))u_2 + X(1 + X)s.$$

Both u_1 - and u_2 -coefficients vanish in characteristic 2 ($X^2 + X^2 = 0$ and $(X + X^2) + (X + X^2) = 0$), leaving $fg = X(1 + X)s$. $\square$

This is exactly the case excluded over the base ring by Lemma 2.7: a product of two J -elements landing in the s -layer whose s -coefficient $X(1 + X)$ is *not* divisible by t (indeed $v_t(X(1 + X)) = 0$). The obstruction fails here because its proof reduces modulo t to $D[X]/tD[X] = \mathbb{F}_2[X]$, which is not a field.

Proposition 2.10. *For every $q \geq 2$, the sequence $f, g, \underbrace{t, \dots, t}_q$ is an irredundant zero product of length $q + 2$ in $A_q[X]$. Consequently (0) is not $(q + 1)$ -absorbing in $A_q[X]$, and $\omega_{A_q[X]}((0)) \geq q + 2$.*

Proof. By Lemma 2.9, the total product is $fg t^q = X(1 + X)t^q s = X(1 + X) \cdot 0 = 0$. Deleting one copy of t leaves $fg t^{q-1} = X(1 + X)t^{q-1}s$, which is nonzero: in $(D/t^q D)[X]$ s the coefficient $X(1 + X)t^{q-1}$ is nonzero because $t^{q-1} \notin t^q D$ and $X(1 + X) \neq 0$ in $\mathbb{F}_2[X]$. Deleting f leaves

$$g t^q = (1 + X)t^q e_1 + (X + X^2)t^q e_2 + X^2 t^q e_3 \neq 0,$$

since $De_1 \oplus De_2 \oplus De_3$ is D -free and, e.g., the e_1 -coefficient $(1 + X)t^q \neq 0$. Symmetrically, deleting g leaves $f t^q = t^q e_2 + X t^q e_3 \neq 0$. Hence the sequence is irredundant of length $q + 2$.

Setting $n = q + 1$, this exhibits $n + 1 = q + 2$ elements of $A_q[X]$ whose product is 0 while no n of them have product 0; by definition (0) is not n -absorbing in $A_q[X]$. The length bound gives $\omega_{A_q[X]}((0)) = \Omega(A_q[X]) \geq q + 2$ via Proposition 2.4. $\square$

For completeness we compute the value exactly; only the lower bound above is needed for Theorem 2.1.

Proposition 2.11. *For every $q \geq 2$, $\omega_{A_q[X]}((0)) = q + 2$.*

Proof. The lower bound is Proposition 2.10. For the upper bound we repeat the case analysis of Proposition 2.8 verbatim with the base ring D replaced by $B = D[X]$, using $J[X]^3 = 0$, $J[X]^2 = W[X]$, $A_q[X]/J[X] \cong D[X]$ (a domain), and Lemma 2.5 for $B = D[X]$ (valid since $D[X]/tD[X] = \mathbb{F}_2[X]$ is a domain). Let $F_1 \cdots F_m = 0$ be an irredundant zero product in $A_q[X]$ and let h be the number of factors in $J[X]$.

If $h \geq 3$: as before $m \leq 3 \leq q + 2$.

If $h = 1$: the torsion-free argument gives the single $J[X]$ -factor inside $W[X]$, of the form $a(X)s$ with $a(X) \in (D/t^q D)[X]$, and one obtains $m \leq q + 1 \leq q + 2$ exactly as in Proposition 2.8 (the obstruction was not used in that case).

If $h = 2$: let the two $J[X]$ -factors be ϕ, ψ and put $z = \phi\psi \in W[X]$. As in (2.6) the total product is $(\prod_i D_i)z$ with $D_i \in D[X] \setminus \{0\}$ the leading components of the remaining $r = m - 2$ factors, and irredundancy forces $z \neq 0$ with vanishing u_1, u_2 -parts, so $z = a(X)s$ with $a(X) \neq 0$. The one and only difference from Proposition 2.8 is that Lemma 2.7 is unavailable over $D[X]$, so now $v := v_t(a(X)) \geq 0$ may equal 0—and indeed $v = 0$ is realized by $z = fg = X(1 + X)s$ of Lemma 2.9. Writing $a(X) = t^v \tilde{a}(X)$ with $\tilde{a}$ of t -valuation 0, additivity of v_t on $D[X]$ gives, exactly as in (2.7), $(0 :_{D[X]} a(X)s) = t^{q-v} D[X]$. Lemma 2.5 then yields $r \leq q - v \leq q$, so $m = 2 + r \leq q + 2$.

In all cases $m \leq q + 2$, so $\omega_{A_q[X]}((0)) \leq q + 2$, and equality holds. $\square$

2.6 Resolution of Problem 30(c)

Proof of Theorem 2.1. Let $n \geq 3$ and put $q = n - 1 \geq 2$. By Proposition 2.8, $\omega_{A_q}((0)) = q + 1 = n$, so (0) is n -absorbing in A_q (Proposition 2.4). By Proposition 2.10, (0) is not n -absorbing in $A_q[X]$; equivalently, by Proposition 2.11, $\omega_{A_q[X]}((0)) = q + 2 = n + 1 > n$. The ring A_q is Noetherian (Lemma 2.6). Taking $A = A_{n-1}$ and $I = (0)$ proves the displayed statement. By Lemma 2.2, $I = (0)$ is an n -absorbing ideal of A whose extension $I[X] = (0)$ is not n -absorbing in $A[X]$, so Problem 30c, and the equality (1.1), fail for n . As $n \geq 3$ was arbitrary they fail for all $n \geq 3$; the affirmative cases $n \leq 2$ of [1] show this is sharp. $\square$

Example 2.12 (The smallest counterexample). Take $q = 2$, so $n = 3$. Then A_2 is the 7-generator $\mathbb{F}_2[t]$ -algebra of Section 2.2 with $t^2 s = 0$, and $\omega_{A_2}((0)) = 3$, $\omega_{A_2[X]}((0)) = 4$. Thus (0) is a 3-absorbing ideal of A_2 , but (0) is not 3-absorbing in $A_2[X]$: with $f = e_2 + X e_3$ and $g = (1 + X)e_1 + (X + X^2)e_2 + X^2 e_3$,

$$f g t t = 0, \quad \text{while} \quad f g t \neq 0, \quad f t^2 \neq 0, \quad g t^2 \neq 0,$$

so the product of four elements vanishes with no 3-fold subproduct vanishing.

Remark 2.13 (A Noetherian local counterexample). Localizing A_q at the maximal ideal $\mathfrak{m} = (t, e_1, e_2, e_3, u_1, u_2, s)$ replaces the base $D = \mathbb{F}_2[t]$ by the discrete valuation ring $\tilde{D} = \mathbb{F}_2[t]_{(t)}$ with uniformizer t and residue field $\tilde{D}/t\tilde{D} \cong \mathbb{F}_2$. The construction (2.1)–(2.2) and all of the proofs above used only that D is a domain with $D/tD \cong \mathbb{F}_2$ a field and that v_t is additive on D and on $D[X]$ —properties shared by $\tilde{D}$ and $\tilde{D}[X]$. Hence the localization $\tilde{A}_q := (A_q)_{\mathfrak{m}}$, a Noetherian local ring with residue field $\mathbb{F}_2$, satisfies the identical conclusions $\omega_{\tilde{A}_q}((0)) = q + 1$ and $\omega_{\tilde{A}_q[X]}((0)) = q + 2$. Thus the counterexample persists in the Noetherian local setting.

Remark 2.14 (Why $n \geq 3$, and why $\mathbb{F}_2$). Two features of the construction are essential and explain its compatibility with the known affirmative cases.

- (a) $n \geq 3$ is forced. The extra factor manufactured in $A_q[X]$ is the copy of t that the obstruction Lemma 2.7 demands over A_q but the cancellation of Lemma 2.9 avoids over $A_q[X]$. This requires room in the socle, i.e. $q \geq 2$, hence $n = q + 1 \geq 3$. For $q = 1$ one has $ts = 0$, the s -layer is one-dimensional, and no extra factor can appear—consistent with the affirmative case $n = 2$ of [1].

- (b) *The finite residue field $\mathbb{F}_2$ is essential.* The proof of Lemma 2.7 uses that the residue field at t is $\mathbb{F}_2$, while the failure over $A_q[X]$ uses that $\mathbb{F}_2[X]$ is not a field. Over a base with infinite residue fields the conjecture (1.1) holds (e.g. algebras over an infinite field), because then an evaluation/root-counting argument transfers n -absorption to the polynomial ring; the present example shows that this mechanism genuinely requires an infinite residue field.

Remark 2.15. Since one always has $\omega_{R[X]}(I[X]) \geq \omega_R(I)$, our examples show that the Anderson–Badawi conjecture (1.1) fails by exactly one: $\omega_{A_q[X]}((0)) = \omega_{A_q}((0)) + 1$ for every $q \geq 2$. We do not know whether the gap $\omega_{R[X]}(I[X]) - \omega_R(I)$ can exceed 1, nor whether it can be made arbitrarily large; these seem natural follow-up questions.

2.7 Computational verification

All ring-theoretic facts about A_q that reduce to finite checks were verified by direct symbolic computation over $\mathbb{F}_2$, independently of the hand proofs above: (i) the multiplication (2.2) extends to a commutative, associative, unital product on the 7-generator module (2.1) with $J^2 = W$ and $J^3 = 0$ (checked on all basis triples and on random elements with polynomial coefficients in t); (ii) the obstruction of Lemma 2.7, namely that no $(a, b, c, a', b', c') \in \mathbb{F}_2^6$ yields a product (2.3) with vanishing u_1, u_2 -coefficients and s -coefficient 1 (all 64 tuples); and (iii) the identity $fg = X(1 + X)s$ of Lemma 2.9. These checks are performed by the script `scripts/verify_nabs.py` accompanying this paper; the two absorbing-number bounds themselves are established by the structural arguments above.

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